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Exercise Sheet 03

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Due: May 26, 2025
Total points: 12

Please upload your solutions to WueCampus as a scanned document (image format or pdf), a typesetted PDF document, and/or as a `jupyter` notebook.

1. Propositional Logic

(a) Evaluate this complex propositions by recursively applying semantics of logical connectives: **1P**

$$(X_0 \implies X_1) \vee X_2$$

(b) Prove De Morgan's Law: **2P**

a) negation of a disjunction is conjunction of the negations

$$\neg(X_1 \vee X_2) \equiv \neg X_1 \wedge \neg X_2$$

b) negation of a conjunction is disjunction of the negations

$$\neg(X_1 \wedge X_2) \equiv \neg X_1 \vee \neg X_2$$

(c) Find a boolean function F in CNF and DNF which has the following truth table: **2P**

A	B	C	F
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	0

(d) Use The DPLL algorithm to determine whether the following two formulas are satisfiable. If so, find a satisfying assignment in propositional logic. **2P**

a) $(X_1 \vee \neg X_2) \wedge (\neg X_1 \vee \neg X_2 \vee X_3) \wedge (X_2 \vee \neg X_3 \vee X_4) \wedge (\neg X_1 \vee X_3 \vee X_4) \wedge (X_2 \vee \neg X_4)$

b) $(X_1 \vee X_2 \vee X_3) \wedge (\neg X_1 \vee X_2) \wedge (X_1 \vee \neg X_3) \wedge \neg X_1 \wedge (\neg X_2 \vee X_3)$

2. Boolean Algebra

(a) Boolean algebra satisfies some of the laws of arithmetic operations. Prove the following: **3P**

a) $\wedge$ and $\vee$ are **associative**, i.e. $X_1 \wedge (X_2 \wedge X_3) = (X_1 \wedge X_2) \wedge X_3$ and $X_1 \vee (X_2 \vee X_3) = (X_1 \vee X_2) \vee X_3$

b) $\wedge$ is **distributive** over $\vee$, i.e. $X_1 \wedge (X_2 \vee X_3) = (X_1 \wedge X_2) \vee (X_1 \wedge X_3)$

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(b) Use (i) a Karnaugh-Veitch map and (ii) the Quine-McCluskey algorithm to minimize the following Boolean function. 2P

X_3	X_2	X_1	X_0	F
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	1
0	1	0	1	1
0	1	1	0	1
0	1	1	1	1
1	0	0	0	1
1	0	0	1	0
1	0	1	0	1
1	0	1	1	0
1	1	0	0	1
1	1	0	1	0
1	1	1	0	1
1	1	1	1	0